Plan the electric car

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use circuit and mechanical evidence to plan a controlled motor-car test.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 14.*
Faster flux change or more turns increases emf magnitude. Stronger field/effective area changes are valid with suitable controls.

02. Worked example: explain the model
Circuit shows electrical connections; placement plan shows physical clearances. They answer different design questions.

03. Read and interpret the model
All four parts form one series loop; any break stops steady current.

04. Apply the physics
The supplied average speed is 4/4 = 1 m/s. The student's actual distances and times must be measured.

05. Explain your reasoning
No; confounded changes. Keep battery and alignment matched while varying wheel choice, use repeated runs.

06. Review: connect an earlier idea *Review: Unit 7, Day 5.*
$Q=It=24$ C.

07. Exit check
Teacher circuit check, secured components, clear rotating parts; measure distance/time or other approved quantity.

08. STRETCH: test the model
Either side of motor can interrupt current; accessibility, insulation and wire routing guide placement.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 14.*
Emf is 3 times as large. Same turn count and flux change per turn; compare average emf over each interval.

02. Worked example: explain the model
Circuit shows electrical connections; placement plan shows physical clearances. They answer different design questions.

03. Read and interpret the model
All four parts form one series loop; any break stops steady current.

04. Apply the physics
1.5 W; 9 J; 0.833 m/s.

05. Explain your reasoning
No; confounded changes. Keep battery and alignment matched while varying wheel choice, use repeated runs.

06. Review: connect an earlier idea *Review: Unit 7, Day 5.*
Circuit response involves propagation of electromagnetic changes and local charge motion. Drift speed is not the propagation speed of the signal.

07. Exit check
Motor/wire heating, rolling friction, sound, wheel rotational energy and drag; accept relevant terms.

08. STRETCH: test the model
4.8 versus 14.4 W. Use brief teacher-approved observation, remove power promptly, avoid heat buildup and use specified current protection.